MATH 231 – Linear Algebra I

Homework 4 ▪ Transformations, Determinants, Systems and Least Squares

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Due: to be announced.

Matrix Unit	Sections	Topics
Part 3	§15–§21	a matrix from what it does; scaling, translation, shear; orthogonal matrices
Part 4	§22–§25	the 2×2 determinant and area; the rotation matrix; rotations of $\mathbb{R}^3$
Part 5	§26–§29	projection matrices; reflections; the Householder reflector; the direct sum
Part 6	§30–§34	the null space; rank–nullity; rank($\mathbf{A}^\mathsf{T}\mathbf{A}$); the 2×2 inverse
Part 7	§35–§39	cofactors; the Laplace expansion; principal minors; the laws D1–D10
Part 8	§40–§45	row operations; Gaussian and Gauss–Jordan elimination; consistency
Part 9	§46–§49	least squares; the normal equations; fitting a line
Part 10	§50–§54	Cramer's rule; the inverse; the adjugate; the pseudo-inverse
Part 11	§55–§58	Gram–Schmidt; the QR decomposition; QR and least squares
Lecture	—	linear maps and operators; $C^\infty(\mathbb{R})$; affine maps and their matrix $[\,\mathbf{A} : \mathbf{b}\,]$

Grading. Every problem is worth the same number of points.

Section numbers and notation. A bare section number such as §28.4 refers to the matrix unit, whose numbering runs straight through Parts 1–11. $\mathbf{e}_i$ is the standard basis vector with a 1 in slot i. For a matrix $\mathbf{A}$, col[$\mathbf{A}$], row[$\mathbf{A}$] and null[$\mathbf{A}$] are its column space, row space and null space.

Rows and columns. A vector $\mathbf{x} \in \mathbb{R}^n$ is a *column*, an $n \times 1$ matrix. A *row* vector is a $1 \times n$ matrix, an element of $\mathbb{R}^{1\times n}$. A bracket written across the line is a row, so a column written across the line carries a transpose: $\mathbf{x} = [\,x_1, \ldots, x_n\,]^\mathsf{T} \in \mathbb{R}^n$. For example, $[\,3, 1\,] \in \mathbb{R}^{1\times 2}$ is a row, while $[\,3, 1\,]^\mathsf{T} = \left[\begin{smallmatrix} 3 \\ 1 \end{smallmatrix}\right] \in \mathbb{R}^2$ is a column, and $\mathbf{u}^\mathsf{T}\mathbf{v}$ is a number while $\mathbf{u}\mathbf{v}^\mathsf{T}$ is a matrix. Keep the two apart in your answers too: a column written across the line needs its $^\mathsf{T}$.

On the proof problems (A.1(e), A.3(a), (b), A.4(a), (b), (d)–(f), B.2(a)–(c), B.3(b), C.3(b), (c), D.2(a)–(c), E.3(d), (e), F.2(a), and G.2(a)–(c)). None of them requires a clever idea. Write down what you are assuming, justify each step by naming the definition, law or theorem that licenses it, and end with a sentence saying what you have shown. A chain of equalities with no words is not a proof; neither is a paragraph of words with no computation.

Collaboration is encouraged — work at a board together, argue about the problems. What you submit must be written by you, in your own words.

What to submit

Push your written work for Parts A–G to your own MATH 231 GitHub repository, as a single PDF named `hw4.pdf` in a folder named `homework/hw4/`. Handwritten and scanned or photographed is completely fine, as long as it is legible and the pages are in order.

PDF is the only accepted format. A `.docx`, a `.jpg`, a `.png`, a `.heic`, or a link to a file hosted somewhere else will not be graded.

Type Parts A–G up instead and you earn +5 points of extra credit on this assignment. Any typesetting system counts — LaTeX, Word's equation editor, or anything else that produces readable mathematics — and the extra credit is for the whole of Parts A–G, not for typing up some of it. Sketches may be drawn by hand and scanned into the PDF.

Label every answer

Label each answer with the number of the problem it answers — `A.3(b)`, `E.2(d)` — and put your answers **in the same order as they appear in this assignment**. A grader who cannot tell which question a piece of work belongs to cannot award it credit. You do not have to copy the question out: write the label, then your work.

Part A. Matrices that move the plane; linear and affine maps

Matrix Unit, §15–§21 and §24–§25. Identify the point (x, y) with the column $[x, y]^{\mathsf{T}}$. A matrix $\mathbf{M} \in \mathbb{R}^{2\times 2}$ moves each point $\mathbf{p}$ to $\mathbf{Mp}$, and its columns are the points to which it moves $\mathbf{e}_1$ and $\mathbf{e}_2$ (the recipe of §15.1). To move a polygon, list its corners as the columns of a matrix $\mathbf{V} \in \mathbb{R}^{2\times k}$, compute the single product $\mathbf{MV}$, and join the moved corners in the same order. Doing $\mathbf{B}$ first and then $\mathbf{A}$ is the single matrix $\mathbf{AB}$ (§15.4). A square matrix $\mathbf{Q}$ is **orthogonal** if $\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{QQ}^{\mathsf{T}} = \mathbf{I}$ (§20.6), and the 2×2 orthogonal matrices are exactly the rotations $\left[\begin{smallmatrix} a & -c \\ c & a \end{smallmatrix}\right]$ and the reflections $\left[\begin{smallmatrix} a & c \\ c & -a \end{smallmatrix}\right]$ with $a^2 + c^2 = 1$ (§21.2). In the angle form of §24.2 and §24.7,

$$\mathbf{R}_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \qquad \mathbf{F}_\theta = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}:$$

$\mathbf{R}_\theta$ rotates counterclockwise by θ, and $\mathbf{F}_\theta$ reflects across the line through the origin at angle $\theta/2$. Two shapes are used below: the unit square S and the standard triangle Δ, with corner matrices $\mathbf{V}_S = \left[\begin{smallmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{smallmatrix}\right]$ and $\mathbf{V}_\Delta = \left[\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right]$. Draw every sketch on graph paper, or on hand-drawn axes with the grid marked, with the original shape and the moved shape on the same axes.

Problem A.1 (A gallery of motions)**.**
Here are six matrices:

$$\mathbf{M}_1 = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad \mathbf{M}_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{M}_3 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix},$$

$$\mathbf{M}_4 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{M}_5 = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{M}_6 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

(a) For each $\mathbf{M}_i$, compute $\mathbf{M}_i\mathbf{V}_S$, sketch S and its image, and say what $\mathbf{M}_i$ does to the square: a stretch (by how much in each direction), a reflection (across which line), a rotation (by which angle, and which way), a shear (which way), or a collapse onto a line (which line).

(b) For each $\mathbf{M}_i$, read the area of the image of S off your sketch, and compute $\det \mathbf{M}_i$. Check that the area is $|\det \mathbf{M}_i|$ (§23.4). Which of the six reverse orientation?

(c) Which of the six are orthogonal? Write each one that is as $\mathbf{R}_\theta$ or $\mathbf{F}_\theta$ for a specific angle θ.

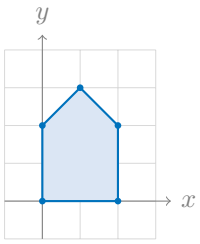
(d) Find the matrix that moves Δ onto the triangle with corners $(0,0)$, $(3,1)$, $(1,2)$, and the one that moves Δ onto the triangle with corners $(0,0)$, $(-1,2)$, $(4,0)$. Use a determinant to find the area of each of the two triangles.

(e) Prove that no 2×2 matrix moves Δ onto the triangle with corners $(1,1)$, $(2,1)$, $(1,2)$. Then write a 3×3 matrix, in homogeneous coordinates (§17.3), that does.

Problem A.2 (One motion after another)**.**
The house below has corners $(0,0)$, $(2,0)$, $(2,2)$, $(1,3)$, $(0,2)$, taken in that order. Let

$$\mathbf{V}_H = \begin{bmatrix} 0 & 2 & 2 & 1 & 0 \\ 0 & 0 & 2 & 3 & 2 \end{bmatrix}, \qquad \mathbf{R} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \qquad \mathbf{F} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix},$$

so that $\mathbf{R}$ is the quarter turn counterclockwise and $\mathbf{F}$ is the reflection across the x-axis.



(a) Compute $\mathbf{R}\mathbf{V}_H$ and sketch the turned house.

(b) Compute $\mathbf{F}(\mathbf{R}\mathbf{V}_H)$, which turns the house and then reflects it, and $\mathbf{R}(\mathbf{F}\mathbf{V}_H)$, which reflects it and then turns it. Sketch both. Do they agree?

(c) Compute the single matrices $\mathbf{FR}$ and $\mathbf{RF}$, and check that $(\mathbf{FR})\mathbf{V}_H = \mathbf{F}(\mathbf{R}\mathbf{V}_H)$. Each of $\mathbf{FR}$ and $\mathbf{RF}$ is a reflection: write each as $\mathbf{F}_\theta$, and name the line it reflects across.

(d) Since $\mathbf{R} = \mathbf{R}_{\pi/2}$, use §24.6 to write $\mathbf{R}^2$, $\mathbf{R}^3$ and $\mathbf{R}^4$ as rotations $\mathbf{R}_\theta$ without multiplying. Then check your answer for $\mathbf{R}^2$ by multiplying.

(e) Write the 3×3 matrix, in homogeneous coordinates (§17.3–§17.4), that first turns the house by $\mathbf{R}$ and then translates it by $[\,3,1\,]^\mathsf{T}$. Apply it to all five corners with one matrix product, and sketch the result.

Problem A.3 (Orthogonal matrices)**.**

(a) Let $\mathbf{Q}_1$ and $\mathbf{Q}_2$ be orthogonal $n \times n$ matrices. Prove that $\mathbf{Q}_1\mathbf{Q}_2$ is orthogonal, naming the laws of §8.4 you use.

(b) Using the angle-addition formulas $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$ and $\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$, prove that $\mathbf{F}_\alpha\mathbf{F}_\beta = \mathbf{R}_{\alpha-\beta}$: two reflections make a rotation. Check it for $\alpha = \pi/2$ and $\beta = 0$ by multiplying.

(c) Find every 2×2 orthogonal matrix whose first column is $[\,\frac{5}{13}, \frac{12}{13}\,]^{\mathsf{T}}$. For the rotation, give $\cos\theta$ and $\sin\theta$. For the reflection, find a nonzero vector on its mirror line and check that the matrix leaves it where it is.

(d) Let $\mathbf{M} = \mathbf{R}_x(\frac{\pi}{2})\,\mathbf{R}_z(\frac{\pi}{2})$, with

$$\mathbf{R}_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}, \qquad \mathbf{R}_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

the rotations of §25.1–§25.2. Compute $\mathbf{M}$ and $\mathbf{M}^{\mathsf{T}}\mathbf{M}$. Find every vector $\mathbf{v} \in \mathbb{R}^3$ with $\mathbf{M}\mathbf{v} = \mathbf{v}$, and compute $\mathbf{M}^3$. ($\mathbf{M}$ is the rotation of space by $2\pi/3$ about the line you found.)

Problem A.4 (Linear and affine maps)**.**

From the lecture on linear and affine maps. Functions are added and scaled point by point: $(f+g)(x) = f(x) + g(x)$ and $(\alpha f)(x) = \alpha f(x)$. $C^\infty(\mathbb{R})$ is the set of functions $f : \mathbb{R} \to \mathbb{R}$ that have derivatives of every order, $f', f'', f''', \ldots$. A map T whose inputs are vectors of $\mathbb{R}^n$, or functions in $C^\infty(\mathbb{R})$, is **linear** if $T(u+v) = T(u) + T(v)$ and $T(\alpha u) = \alpha\,T(u)$ for all inputs u, v and every number α. For $\mathbf{A} \in \mathbb{R}^{2\times 2}$ and $\mathbf{b} \in \mathbb{R}^2$, the **affine map** $T(\mathbf{p}) = \mathbf{A}\mathbf{p} + \mathbf{b}$ applies $\mathbf{A}$ to the point $\mathbf{p}$ and then shifts it by $\mathbf{b}$. Its matrix is the 2×3 matrix $[\,\mathbf{A} : \mathbf{b}\,]$, whose first two columns are those of $\mathbf{A}$ and whose third column is $\mathbf{b}$:

$$[\,\mathbf{A} : \mathbf{b}\,]\begin{bmatrix} x \\ y \\ t \end{bmatrix} = \mathbf{A}\begin{bmatrix} x \\ y \end{bmatrix} + t\,\mathbf{b}, \qquad \text{so} \qquad T\big([\,x, y\,]^{\mathsf{T}}\big) = [\,\mathbf{A} : \mathbf{b}\,]\,[\,x, y, 1\,]^{\mathsf{T}}.$$

(a) Let $L : C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R})$ be $L(f) = f'' - 3f' + 2f$. Prove that L is linear, naming the rules of differentiation you use. Compute $L(e^x)$, $L(e^{2x})$ and $L(x)$, and then use linearity alone, with no more differentiation, to find $L(5e^x - 2e^{2x} + 4x)$.

(b) Prove that every linear map T sends $\mathbf{0}$ to $\mathbf{0}$. Then prove that none of these three maps is linear, by finding specific inputs at which one of the two properties fails: $g : \mathbb{R} \to \mathbb{R}$, $g(x) = 3x + 2$; $h : \mathbb{R}^2 \to \mathbb{R}$, $h\big([\,x, y\,]^{\mathsf{T}}\big) = xy$; and $N : C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R})$, $N(f) = f\,f'$. (h and N do send $\mathbf{0}$ to $\mathbf{0}$, so for them that test is not enough.)

(c) Let $\mathbf{A} = \left[\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right]$, $\mathbf{b} = [\,2, -1\,]^{\mathsf{T}}$ and $T(\mathbf{p}) = \mathbf{A}\mathbf{p} + \mathbf{b}$. Write $[\,\mathbf{A} : \mathbf{b}\,]$. Move the unit square S with one product: append a row of 1s to $\mathbf{V}_S$, multiply by $[\,\mathbf{A} : \mathbf{b}\,]$ on the left, and sketch S and $T(S)$. Compare the area of $T(S)$ with $|\det\mathbf{A}|$.

(d) Prove that an affine map $T(\mathbf{p}) = \mathbf{A}\mathbf{p} + \mathbf{b}$ is linear if and only if $\mathbf{b} = \mathbf{0}$. Then prove that the map $\mathbb{R}^3 \to \mathbb{R}^2$, $\mathbf{u} \mapsto [\,\mathbf{A} : \mathbf{b}\,]\,\mathbf{u}$, is linear for every $\mathbf{A}$ and $\mathbf{b}$, naming the laws of §8.4 you use. (So T is a linear map one dimension up, used only on the inputs with $t = 1$; compare the remark in §17.3.)

(e) Let $S(\mathbf{p}) = \mathbf{C}\mathbf{p} + \mathbf{d}$ be a second affine map. Prove that doing T and then S is the affine map with matrix $[\,\mathbf{C}\mathbf{A} : \mathbf{C}\mathbf{b} + \mathbf{d}\,]$. The product $[\,\mathbf{C} : \mathbf{d}\,][\,\mathbf{A} : \mathbf{b}\,]$ of two 2×3 matrices is not defined; the 3×3 matrices of §17.4 fix this. For the T of (c) and $\mathbf{C} = \left[\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right]$, $\mathbf{d} = [0, 1]^{\mathsf{T}}$, check that $\left[\begin{smallmatrix} \mathbf{C} & \mathbf{d} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{smallmatrix}\right] \left[\begin{smallmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{smallmatrix}\right]$ has $[\,\mathbf{C}\mathbf{A} : \mathbf{C}\mathbf{b} + \mathbf{d}\,]$ as its top two rows.

(f) For the T of (c), find the point $\mathbf{c}$ with $T(\mathbf{c}) = \mathbf{c}$, and prove that $T(\mathbf{p}) = \mathbf{c} + \mathbf{A}(\mathbf{p} - \mathbf{c})$ for every $\mathbf{p} \in \mathbb{R}^2$. Check that $\mathbf{A} = \sqrt{2}\,\mathbf{R}_{\pi/4}$: T turns the plane by $\pi/4$ about $\mathbf{c}$ and stretches every distance from $\mathbf{c}$ by $\sqrt{2}$.

Part B. Projections, reflections, and the direct sum

Matrix Unit, §26–§29. For a nonzero $\mathbf{u} \in \mathbb{R}^n$, the projection matrix onto the line $\operatorname{span}\{\mathbf{u}\}$ is $\mathbf{P}_{\mathbf{u}} = \mathbf{u}\mathbf{u}^{\mathsf{T}}/\mathbf{u}^{\mathsf{T}}\mathbf{u}$, and $\mathbf{P}_{\mathbf{u}}\mathbf{v} = \operatorname{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\mathbf{u}^{\mathsf{T}}\mathbf{v}}{\mathbf{u}^{\mathsf{T}}\mathbf{u}}\,\mathbf{u}$ (§26.2). A square matrix $\mathbf{P}$ is a **projection matrix** if $\mathbf{P}^2 = \mathbf{P}$ (§27.2), and an **orthogonal projection matrix** if also $\mathbf{P}^{\mathsf{T}} = \mathbf{P}$ (§27.4). By the theorem of §27.4, for an orthogonal projection matrix the leftover $\mathbf{v} - \mathbf{P}\mathbf{v}$ is orthogonal to every vector of $\operatorname{col}[\mathbf{P}]$. The reflection across $\operatorname{span}\{\mathbf{u}\}$ is $2\mathbf{P}_{\mathbf{u}} - \mathbf{I}$ (§28.4). For $\mathbf{u} \in \mathbb{R}^2$ not on the negative x-axis, the **Householder reflector** is

$$\mathbf{H} = 2\mathbf{v}\mathbf{v}^{\mathsf{T}} - \mathbf{I}_2, \qquad \mathbf{v} = \frac{\mathbf{m}}{\|\mathbf{m}\|}, \qquad \mathbf{m} = \tfrac{1}{2}\big(\mathbf{u} + \|\mathbf{u}\|\,\mathbf{e}_1\big),$$

and $\mathbf{H}\mathbf{u} = \|\mathbf{u}\|\,\mathbf{e}_1$ (§28.5–§28.6). If U and W are subspaces of $\mathbb{R}^n$, then $\mathbb{R}^n = U \oplus W$, the **direct sum**, means that every vector of $\mathbb{R}^n$ can be written in exactly one way as $\mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$ and $\mathbf{w} \in W$. By the theorem of §29.3, this holds if and only if every vector is such a sum and $U \cap W = \{\mathbf{0}\}$.

Problem B.1 (Building projection matrices)**.**

(a) Compute $\mathbf{P}_{\mathbf{u}}$, with exact fractions, for $\mathbf{u} = [1, -2]^{\mathsf{T}}$, $[2, -1, 2]^{\mathsf{T}}$ and $[1, 2, 2]^{\mathsf{T}}$. For each, check that $\mathbf{P}_{\mathbf{u}}$ is symmetric and that $\mathbf{P}_{\mathbf{u}}\mathbf{u} = \mathbf{u}$, and give its rank.

(b) With $\mathbf{u} = [1, 2, 2]^{\mathsf{T}}$ and $\mathbf{v} = [3, 0, 3]^{\mathsf{T}}$, compute $\mathbf{P}_{\mathbf{u}}\mathbf{v}$ and check it against the formula for $\operatorname{proj}_{\mathbf{u}}(\mathbf{v})$. Then compute $(\mathbf{I} - \mathbf{P}_{\mathbf{u}})\mathbf{v}$ and check that it is orthogonal to $\mathbf{u}$.

(c) Let $\mathbf{u} = [\cos\theta, \sin\theta]^{\mathsf{T}}$. Show that $\mathbf{P}_{\mathbf{u}} = \begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}$, and that $2\mathbf{P}_{\mathbf{u}} - \mathbf{I} = \mathbf{F}_{2\theta}$, the reflection across the line at angle θ. Use $\cos 2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$ and $\sin 2\theta = 2\sin\theta\cos\theta$.

(d) Which matrix of Problem A.1 is $\mathbf{P}_{\mathbf{u}}$ for $\mathbf{u} = [1, 1]^{\mathsf{T}}$? Which is $2\mathbf{P}_{\mathbf{u}} - \mathbf{I}$ for the same $\mathbf{u}$?

(e) Let $\mathbf{1} = [1, 1, 1, 1]^{\mathsf{T}} \in \mathbb{R}^4$. Write out $\mathbf{P}_{\mathbf{1}}$, and compute $\mathbf{P}_{\mathbf{1}}\mathbf{x}$ and $(\mathbf{I} - \mathbf{P}_{\mathbf{1}})\mathbf{x}$ for $\mathbf{x} = [4, 7, 1, 8]^{\mathsf{T}}$. Then show that for every $\mathbf{x} \in \mathbb{R}^4$, every entry of $\mathbf{P}_{\mathbf{1}}\mathbf{x}$ equals the **mean** $\frac{1}{4}(x_1 + x_2 + x_3 + x_4)$ of the entries of $\mathbf{x}$.

Problem B.2 (Projections that are not orthogonal)**.**
In (a)–(c), $\mathbf{P}$ is an $n \times n$ matrix with $\mathbf{P}^2 = \mathbf{P}$.

(a) Prove that $\mathbf{P}^{\mathsf{T}}$ is also a projection matrix. Name the law of §8.4 you use.

(b) Prove that a vector $\mathbf{v} \in \mathbb{R}^n$ lies in $\operatorname{col}[\mathbf{P}]$ if and only if $\mathbf{P}\mathbf{v} = \mathbf{v}$.

(c) Suppose also that $\mathbf{P}^{\mathsf{T}} = \mathbf{P}$. Prove that $\|\mathbf{Pv}\| \le \|\mathbf{v}\|$ for every $\mathbf{v} \in \mathbb{R}^n$.

Hint. Write $\mathbf{v} = \mathbf{Pv} + (\mathbf{v} - \mathbf{Pv})$, and use the theorem of §27.4 and the Pythagorean identity.

(d) Let $\mathbf{P} = \left[\begin{smallmatrix} 1 & 2 \\ 0 & 0 \end{smallmatrix}\right]$. Check that $\mathbf{P}^2 = \mathbf{P}$ and that $\mathbf{P}$ is not symmetric. Describe $\operatorname{col}[\mathbf{P}]$, and the set of vectors that $\mathbf{P}$ sends to $\mathbf{0}$, as lines. Compare $\|\mathbf{Pe}_2\|$ with $\|\mathbf{e}_2\|$.

(e) With the same $\mathbf{P}$, split $\mathbf{x} = [\,3, 1\,]^{\mathsf{T}}$ as $\mathbf{Px} + (\mathbf{I} - \mathbf{P})\mathbf{x}$ (§29.5). Are the two pieces orthogonal? Is $\|\mathbf{x}\|^2 = \|\mathbf{Px}\|^2 + \|(\mathbf{I} - \mathbf{P})\mathbf{x}\|^2$?

Problem B.3 (The Householder reflector, and direct sums)**.**

(a) Let $\mathbf{u} = [\,8, 15\,]^{\mathsf{T}}$. Compute $\mathbf{m}$, $\mathbf{v}$ and $\mathbf{H} = 2\mathbf{v}\mathbf{v}^{\mathsf{T}} - \mathbf{I}$ with exact entries. Check that $\mathbf{Hu} = [\,17, 0\,]^{\mathsf{T}}$, that the first column of $\mathbf{H}$ is $\mathbf{u}/\|\mathbf{u}\|$, and that $\mathbf{H}^2 = \mathbf{I}$.

(b) Let $\mathbf{u} = r\,[\,\cos\theta, \sin\theta\,]^{\mathsf{T}}$ with $r > 0$ and $-\pi < \theta < \pi$. Prove that the Householder reflector of $\mathbf{u}$ is $\mathbf{F}_\theta$, so its mirror is the line at angle $\theta/2$, halfway between $\mathbf{u}$ and the positive x-axis.

Hint. The half-angle identities $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$ and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$ give $\mathbf{m} = r\cos\frac{\theta}{2}\,\mathbf{w}$, where $\mathbf{w} = [\,\cos\frac{\theta}{2}, \sin\frac{\theta}{2}\,]^{\mathsf{T}}$. Then use B.1(c).

(c) Let $U = \operatorname{span}\{[\,1, 0\,]^{\mathsf{T}}\}$ and $W = \operatorname{span}\{[\,1, 2\,]^{\mathsf{T}}\}$, subspaces of $\mathbb{R}^2$. Use the theorem of §29.3 to show that $\mathbb{R}^2 = U \oplus W$, and split $\mathbf{x} = [\,4, 6\,]^{\mathsf{T}}$ into its piece in U and its piece in W. Find the 2×2 matrix $\mathbf{P}$ that sends every $\mathbf{x}$ to its piece in U, check that $\mathbf{P}^2 = \mathbf{P}$, and say whether $\mathbf{P}$ is an orthogonal projection matrix.

(d) Let $W = \{\mathbf{x} \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$, a subspace of $\mathbb{R}^3$. Show that $\mathbb{R}^3 = \operatorname{span}\{[\,1, 1, 0\,]^{\mathsf{T}}\} \oplus W$, and split $[\,3, 1, 2\,]^{\mathsf{T}}$ into its two pieces. Then show that $\mathbb{R}^3$ is *not* $\operatorname{span}\{[\,1, -1, 0\,]^{\mathsf{T}}\} \oplus W$.

Part C. Determinants

Matrix Unit, §22–§23 and §35–§39. The parallelogram with sides the columns of $\mathbf{A} \in \mathbb{R}^{2\times 2}$ has area $|\det \mathbf{A}|$ (§23.4), and $\mathbf{A}$ multiplies the area of every region by $|\det \mathbf{A}|$ (§23.6). For $\mathbf{A} \in \mathbb{R}^{n\times n}$, the minor M_{ij} is the determinant of $\mathbf{A}$ with row i and column j deleted, and the cofactor is $C_{ij} = (-1)^{i+j} M_{ij}$ (§35). Then $\det \mathbf{A} = \sum_j a_{1j} C_{1j}$, and expanding along any other row or column gives the same number (§36).
The laws of §38.1 are:

D1	$\det \mathbf{I}_n = 1$	**D6**	two equal rows give $\det \mathbf{A} = 0$
D2	$\det \mathbf{A}^{\mathsf{T}} = \det \mathbf{A}$	**D7**	det is linear in each row separately
D3	a zero row gives $\det \mathbf{A} = 0$	**D8**	adding a multiple of one row to another leaves det unchanged
D4	scaling one row by α scales det by α; so $\det(\alpha\mathbf{A}) = \alpha^n \det \mathbf{A}$	**D9**	$\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$
D5	swapping two rows changes the sign	**D10**	$\det \mathbf{A} \neq 0 \iff \operatorname{rank} \mathbf{A} = n \iff \operatorname{null}[\mathbf{A}] = \{\mathbf{0}\}$

By D2, every law about rows is also a law about columns. The determinant of a triangular matrix is the product of its diagonal entries (§39.1). Keeping the rows *and* the columns numbered by a set $S \subseteq \{1, \dots, n\}$ gives a principal submatrix, whose determinant is a **principal minor**; for $S = \{1, \dots, k\}$ it is the k-th **leading principal minor** Δ_k (§37.1). A symmetric $\mathbf{A}$ is **positive definite** if $\mathbf{x}^{\mathsf{T}}\mathbf{Ax} > 0$ for every $\mathbf{x} \neq \mathbf{0}$, which happens exactly when $\Delta_1, \dots, \Delta_n$ are all positive (Sylvester's criterion, §37.3).

Problem C.1 (Area)**.**

(a) Find the area of the parallelogram with sides $[\,4,1\,]^{\mathsf{T}}$ and $[\,2,5\,]^{\mathsf{T}}$, and of the triangle with corners $(1,1)$, $(5,2)$ and $(3,6)$.

(b) Find every number k for which the triangle with corners $(0,0)$, $(3,1)$ and $(k,2)$ has area 4.

(c) A region of the plane has area 5. Let $\mathbf{M} = \left[\begin{smallmatrix} 3 & 1 \\ 2 & 4 \end{smallmatrix}\right]$. Find the area of its image under $\mathbf{M}$, under $\mathbf{M}^2$, under $\mathbf{R}_{\pi/3}\mathbf{M}$, and under $\frac{1}{2}\mathbf{M}$, naming the facts of §22 and §23.6 you use.

Problem C.2 (Computing determinants)**.**

Let

$$\mathbf{B} = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 4 & 0 \\ -2 & 1 & 5 \end{bmatrix}, \qquad \mathbf{A} = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 3 & 1 & 0 & 2 \\ 0 & 0 & 4 & 0 \\ 2 & 0 & 1 & 3 \end{bmatrix}, \qquad \mathbf{T}_k = \begin{bmatrix} 1 & k & 0 \\ k & 1 & k \\ 0 & k & 1 \end{bmatrix}.$$

(a) Compute $\det \mathbf{B}$ by expanding along the first row, and again by expanding down the third column.

(b) Compute $\det \mathbf{A}$, expanding along whichever row or column makes the work shortest (§36.4).

(c) Compute $\det \mathbf{B}$ a third time: reduce $\mathbf{B}$ to an upper triangular matrix by row operations, record the effect of each one on the determinant (§38.3), and finish with §39.1.

(d) Find every number k for which $\det \mathbf{T}_k = 0$. For the positive one, find a nonzero vector $\mathbf{x}$ with $\mathbf{T}_k\mathbf{x} = \mathbf{0}$.

Problem C.3 (Laws, two proofs, and positive definite matrices)**.**

(a) Let $\mathbf{A}$ be 3×3 with $\det \mathbf{A} = 4$. Find $\det(2\mathbf{A})$, $\det(-\mathbf{A})$, $\det(\mathbf{A}^3)$ and $\det(\mathbf{A}^{\mathsf{T}}\mathbf{A})$. Then find the determinant of the matrix obtained from $\mathbf{A}$ by (i) swapping rows 1 and 3; (ii) adding 5 times row 2 to row 1; (iii) tripling row 2 and then swapping rows 1 and 2. Name the law used each time.

(b) A square matrix $\mathbf{K}$ is **skew-symmetric** if $\mathbf{K}^{\mathsf{T}} = -\mathbf{K}$. Prove that if $\mathbf{K}$ is $n \times n$ and skew-symmetric and n is odd, then $\det \mathbf{K} = 0$. Check it on $\left[\begin{smallmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ 2 & -3 & 0 \end{smallmatrix}\right]$, and show that the skew-symmetric $\left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$ has nonzero determinant.

(c) Prove that

$$\det \begin{bmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{bmatrix} = (b-a)(c-a)(c-b),$$

and deduce that this determinant is 0 exactly when two of a, b, c are equal.

Hint. Subtract column 1 from columns 2 and 3 (D8 for columns), expand along the first row, and factor.

(d) Let $\mathbf{S} = \left[\begin{smallmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{smallmatrix}\right]$. Find all seven principal minors of $\mathbf{S}$ and its three leading principal minors. Is $\mathbf{S}$ positive definite?

(e) For $\mathbf{A} = \left[\begin{smallmatrix} 1 & 3 \\ 3 & 4 \end{smallmatrix}\right]$, compute Δ_1 and Δ_2, and find a vector $\mathbf{x}$ with $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} < 0$.

Hint. Look at the vector used in the proof of Sylvester's criterion in §37.3.

Part D. The null space and rank–nullity

Matrix Unit, §30–§34. The **null space** of $\mathbf{A} \in \mathbb{R}^{m\times n}$ is $\operatorname{null}[\mathbf{A}] = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} = \mathbf{0}\}$, a subspace of $\mathbb{R}^n$, and its dimension is the **nullity** (§31). The matrix has **full column rank** if $\operatorname{rank}\mathbf{A} = n$, **full row rank** if $\operatorname{rank}\mathbf{A} = m$, and **full rank** if $\operatorname{rank}\mathbf{A} = \min(m,n)$ (§30.1). The rank–nullity theorem says $\operatorname{rank}\mathbf{A} + \operatorname{nullity}\mathbf{A} = n$ (§32). If $\mathbf{x}_p$ is one solution of $\mathbf{Ax} = \mathbf{b}$, the solutions are exactly $\mathbf{x}_p + \mathbf{z}$ with $\mathbf{z} \in \operatorname{null}[\mathbf{A}]$ (§31.6). If $\mathbf{r}$ is a row of $\mathbf{A}$ and $\mathbf{z} \in \operatorname{null}[\mathbf{A}]$, then $\mathbf{rz} = 0$, so $\mathbf{z}$ is orthogonal to the column $\mathbf{r}^{\mathsf{T}}$ (§31.7), and $\operatorname{rank}(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \operatorname{rank}\mathbf{A} = \operatorname{rank}\mathbf{A}^{\mathsf{T}}$ (§33). For square $\mathbf{A}$, a subspace W is **invariant** under $\mathbf{A}$ if $\mathbf{Aw} \in W$ for every $\mathbf{w} \in W$ (§30.4). If $ad - bc \neq 0$, then $\left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]^{-1} = \frac{1}{ad-bc}\left[\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right]$, and if $ad - bc = 0$ there is no inverse (§34).

Problem D.1 (Null spaces, rank and nullity)**.**
Let

$$\mathbf{A}_1 = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}, \quad \mathbf{A}_2 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}, \quad \mathbf{A}_3 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 0 \end{bmatrix}, \quad \mathbf{A}_4 = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{bmatrix}.$$

(a) For each $\mathbf{A}_i$, write $\operatorname{null}[\mathbf{A}_i]$ as the span of as few vectors as possible, or show that it is $\{\mathbf{0}\}$, and give the nullity.

(b) Give the rank of each $\mathbf{A}_i$ and check rank–nullity. Say whether each has full rank, full column rank, full row rank, or none of these.

(c) Check that $\mathbf{rz} = 0$ for every row $\mathbf{r}$ of $\mathbf{A}_2$ and every vector $\mathbf{z}$ you used to span $\operatorname{null}[\mathbf{A}_2]$.

(d) Find every solution of $\mathbf{A}_2\mathbf{x} = [\,3, 1, 4\,]^{\mathsf{T}}$ in the form $\mathbf{x}_p + \mathbf{z}$, $\mathbf{z} \in \operatorname{null}[\mathbf{A}_2]$. Then show that $\mathbf{A}_2\mathbf{x} = [\,1, 1, 1\,]^{\mathsf{T}}$ has no solution.

(e) A matrix $\mathbf{A} \in \mathbb{R}^{6\times 9}$ has rank 4. Find $\operatorname{nullity}\mathbf{A}$, $\operatorname{rank}\mathbf{A}^{\mathsf{T}}$, $\operatorname{nullity}\mathbf{A}^{\mathsf{T}}$, $\operatorname{rank}(\mathbf{A}^{\mathsf{T}}\mathbf{A})$ and $\operatorname{rank}(\mathbf{A}\mathbf{A}^{\mathsf{T}})$, naming the result you use each time.

Problem D.2 (Proofs, invariant lines, and the 2×2 inverse)**.**

(a) Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{B} \in \mathbb{R}^{p\times m}$. Prove that $\operatorname{null}[\mathbf{A}] \subseteq \operatorname{null}[\mathbf{BA}]$. Then prove that if $\operatorname{null}[\mathbf{B}] = \{\mathbf{0}\}$, then $\operatorname{null}[\mathbf{BA}] = \operatorname{null}[\mathbf{A}]$.

(b) Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{B} \in \mathbb{R}^{n\times p}$. Prove that $\operatorname{col}[\mathbf{AB}] \subseteq \operatorname{col}[\mathbf{A}]$, and deduce that $\operatorname{rank}(\mathbf{AB}) \leq \operatorname{rank}\mathbf{A}$. Then use the first statement of (a), applied to the product $\mathbf{AB}$, together with rank–nullity, to show that $\operatorname{rank}(\mathbf{AB}) \leq \operatorname{rank}\mathbf{B}$.

(c) Let $\mathbf{A}$ be $n \times n$. Prove that $\operatorname{null}[\mathbf{A}]$ and $\operatorname{col}[\mathbf{A}]$ are invariant under $\mathbf{A}$. Then find every line through the origin that is invariant under $\left[\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right]$.

(d) For which numbers k is $\mathbf{B}_k = \left[\begin{smallmatrix} k & 4 \\ 1 & k \end{smallmatrix}\right]$ invertible? For $k = 3$, find $\mathbf{B}_3^{-1}$ and use it to solve $\mathbf{B}_3\mathbf{x} = [\,1, 2\,]^{\mathsf{T}}$. For $k = 2$, find $\operatorname{null}[\mathbf{B}_2]$ and $\operatorname{col}[\mathbf{B}_2]$.

Part E. Elimination and the inverse

Matrix Unit, §40–§45 and §50–§53. The elementary row operations are **(E1)** swap two rows, **(E2)** scale a row by a nonzero number, **(E3)** add a multiple of one row to another. They do not change the solutions of a system (§40.3), and each one is multiplication on the left by the **elementary**

matrix obtained by doing it to $\mathbf{I}$ (§40.4). Row-echelon form is defined in §42.5 and reduced row-echelon form (RREF) in §43.1. By §43.4, the columns of $\mathbf{A}$ in the pivot positions of its RREF form a basis of col[$\mathbf{A}$], and setting one free variable to 1 and the others to 0, for each free variable in turn, gives a basis of null[$\mathbf{A}$]. A system is consistent exactly when elimination produces no row $[\,0\ \cdots\ 0 \mid c\,]$ with $c \neq 0$ (§44.3).

A matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ is **invertible** if there is a $\mathbf{B}$ with $\mathbf{AB} = \mathbf{BA} = \mathbf{I}$ (§51.1). Gauss–Jordan elimination turns $[\,\mathbf{A} \mid \mathbf{I}\,]$ into $[\,\mathbf{I} \mid \mathbf{A}^{-1}\,]$ (§51.4). The **adjugate** $\operatorname{adj}\mathbf{A}$ is the transpose of the matrix of cofactors, and $\mathbf{A}^{-1} = \frac{1}{\det\mathbf{A}}\operatorname{adj}\mathbf{A}$ (§52). If $\det\mathbf{A} \neq 0$, **Cramer's rule** gives the solution of $\mathbf{Ax} = \mathbf{b}$ as $x_i = \det\mathbf{A}_i(\mathbf{b})/\det\mathbf{A}$, where $\mathbf{A}_i(\mathbf{b})$ is $\mathbf{A}$ with column i replaced by $\mathbf{b}$ (§50.1). The laws of §53.1 are **I1** $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$, **I2** $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$, **I3** $(\mathbf{A}^{\mathsf{T}})^{-1} = (\mathbf{A}^{-1})^{\mathsf{T}}$, **I4** $(\alpha\mathbf{A})^{-1} = \alpha^{-1}\mathbf{A}^{-1}$, **I5** $\det(\mathbf{A}^{-1}) = 1/\det\mathbf{A}$ and **I6** $(\mathbf{A}^{k})^{-1} = (\mathbf{A}^{-1})^{k}$.

Problem E.1 (Elimination, step by step)**.**

(a) Solve the system

$$x_1 + 2x_2 + x_3 = 1, \qquad 2x_1 + 5x_2 + 3x_3 = 3, \qquad -x_1 + x_2 + 4x_3 = 6$$

by Gaussian elimination and back substitution, writing down each row operation. Then read off the determinant of the coefficient matrix $\mathbf{A}$ from the pivots (§42.6).

(b) Write the elementary matrix of each row operation you used in (a). Check that their product, in the right order, times $\mathbf{A}$ is the upper triangular matrix you reached.

(c) Solve $\mathbf{Lx} = [\,4, 5, 3\,]^{\mathsf{T}}$ by forward substitution (§41.2), where $\mathbf{L} = \left[\begin{smallmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ -1 & 2 & 1 \end{smallmatrix}\right]$.

Problem E.2 (Reduced row-echelon form, bases, and consistency)**.**
Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 & 1 & -1 \\ 2 & 4 & 1 & 4 & 1 \\ 1 & 2 & 1 & 3 & 2 \end{bmatrix}.$$

(a) Bring $\mathbf{A}$ to reduced row-echelon form, naming each row operation. Mark the pivots, and list the pivot variables and the free variables.

(b) Use §43.4 to give a basis of col[$\mathbf{A}$] made of columns of $\mathbf{A}$, and a basis of null[$\mathbf{A}$]. Give the rank and the nullity, and check rank–nullity.

(c) Write each non-pivot column of $\mathbf{A}$ as a combination of the pivot columns of $\mathbf{A}$, and check one of them by computing.

(d) Find a condition on b_1, b_2, b_3 that holds exactly when $\mathbf{Ax} = [\,b_1, b_2, b_3\,]^{\mathsf{T}}$ is consistent. Then find every solution of $\mathbf{Ax} = [\,2, 5, 3\,]^{\mathsf{T}}$, and show that $\mathbf{Ax} = [\,1, 1, 1\,]^{\mathsf{T}}$ has none.

Problem E.3 (The inverse)**.**
Let

$$\mathbf{M} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 4 & 1 \end{bmatrix}, \qquad \mathbf{N} = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}.$$

(a) Find $\mathbf{M}^{-1}$ by Gauss–Jordan elimination, and check that $\mathbf{MM}^{-1} = \mathbf{I}$.

(b) Find $\mathbf{N}^{-1}$ by the adjugate, and check that $\mathbf{N}\mathbf{N}^{-1} = \mathbf{I}$.

(c) Use Cramer's rule to find x_2 alone in $\mathbf{N}\mathbf{x} = [\,1, 3, 5\,]^{\mathsf{T}}$. Then confirm it with $\mathbf{N}^{-1}$ from (b).

(d) Let $\mathbf{A}, \mathbf{B}, \mathbf{C}$ be invertible $n \times n$ matrices. Prove that $(\mathbf{ABC})^{-1} = \mathbf{C}^{-1}\mathbf{B}^{-1}\mathbf{A}^{-1}$, naming the laws of §8.4 and §53.1 you use. Then prove that if $\mathbf{A}$ is invertible and $\mathbf{AB} = \mathbf{AC}$, then $\mathbf{B} = \mathbf{C}$.

(e) Let $\mathbf{A}$ be $n \times n$ with $\mathbf{A}^2 = \mathbf{0}$. Prove that $\mathbf{A}$ is not invertible, and that $\mathbf{I} - \mathbf{A}$ is invertible with $(\mathbf{I} - \mathbf{A})^{-1} = \mathbf{I} + \mathbf{A}$. Check both statements on $\mathbf{A} = \left[\begin{smallmatrix} 1 & -1 \\ 1 & -1 \end{smallmatrix}\right]$.

Part F. Least squares and the pseudo-inverse

Matrix Unit, §46–§49 and §54. A **least-squares solution** of $\mathbf{A}\mathbf{x} = \mathbf{b}$ is a vector $\hat{\mathbf{x}}$ that makes $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$ as small as possible (§47.1). These are exactly the solutions of the **normal equations** $\mathbf{A}^{\mathsf{T}}\mathbf{A}\,\hat{\mathbf{x}} = \mathbf{A}^{\mathsf{T}}\mathbf{b}$, which say that the residual $\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}$ is orthogonal to every column of $\mathbf{A}$ (§47.2–§47.3). The normal equations always have a solution, and it is unique exactly when $\mathbf{A}$ has full column rank (§49.1–§49.2). To fit the line $y = c_0 + c_1 t$ to data points (t_i, y_i), row i of $\mathbf{A}$ is the row vector $[\,1, t_i\,]$ and $\mathbf{b}$ lists the y_i (§49.4). The **pseudo-inverse** is $\mathbf{A}^{+} = (\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathsf{T}}$ if $\mathbf{A}$ has independent columns (§54.2), and $\mathbf{A}^{+} = \mathbf{A}^{\mathsf{T}}(\mathbf{A}\mathbf{A}^{\mathsf{T}})^{-1}$ if $\mathbf{A}$ has independent rows, in which case $\mathbf{A}^{+}\mathbf{b}$ is the shortest solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ (§54.3). For any $\mathbf{A}$, $\mathbf{A}^{+}$ is the one matrix $\mathbf{X}$ with **(MP1)** $\mathbf{AXA} = \mathbf{A}$, **(MP2)** $\mathbf{XAX} = \mathbf{X}$, **(MP3)** $(\mathbf{AX})^{\mathsf{T}} = \mathbf{AX}$ and **(MP4)** $(\mathbf{XA})^{\mathsf{T}} = \mathbf{XA}$ (§54.4).

Problem F.1 (Fitting a line)**.**
The four data points are $(0, 1)$, $(1, 3)$, $(2, 4)$ and $(3, 4)$.

(a) Write the system $\mathbf{A}\mathbf{x} = \mathbf{b}$, with $\mathbf{x} = [\,c_0, c_1\,]^{\mathsf{T}}$, that says the line $y = c_0 + c_1 t$ passes through all four points. Show by elimination that it has no solution.

(b) Form $\mathbf{A}^{\mathsf{T}}\mathbf{A}$ and $\mathbf{A}^{\mathsf{T}}\mathbf{b}$, solve the normal equations, and write down the best line.

(c) Compute the residual $\mathbf{r} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}}$, check that $\mathbf{A}^{\mathsf{T}}\mathbf{r} = \mathbf{0}$, and find the sum of the squared errors. Sketch the four points, the line, and the four vertical segments whose lengths are the entries of $\mathbf{r}$.

(d) Compute $\mathbf{A}^{+} = (\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathsf{T}}$. Check that $\mathbf{A}^{+}\mathbf{A} = \mathbf{I}_2$ and that $\mathbf{A}^{+}\mathbf{b} = \hat{\mathbf{x}}$.

Problem F.2 (Pythagoras, shortest solutions, and Moore–Penrose)**.**

(a) Let $\hat{\mathbf{x}}$ be a least-squares solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$. Prove that $\|\mathbf{b}\|^2 = \|\mathbf{A}\hat{\mathbf{x}}\|^2 + \|\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}\|^2$. Check it on Problem F.1.

(b) Use §54.3 to find the shortest solution $\mathbf{x}^{+}$ of $x_1 + 2x_2 + 2x_3 = 9$. Check that $\mathbf{x}^{+}$ is orthogonal to every vector of the null space of the 1×3 matrix $[\,1, 2, 2\,]$, and compare its length with that of the solution $[\,9, 0, 0\,]^{\mathsf{T}}$.

(c) The matrix $\mathbf{A}$ below has neither independent columns nor independent rows:

$$\mathbf{A} = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}, \qquad \mathbf{X} = \frac{1}{25}\begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}.$$

Check that $\mathbf{X}$ satisfies MP1–MP4, so $\mathbf{X} = \mathbf{A}^{+}$. Show that $\mathbf{AX}$ and $\mathbf{XA}$ are the projection matrices $\mathbf{P}_{\mathbf{y}}$ and $\mathbf{P}_{\mathbf{z}}$ of §26.2 for $\mathbf{y} = [\,2, 1\,]^{\mathsf{T}}$ and $\mathbf{z} = [\,1, 2\,]^{\mathsf{T}}$.

(d) With $\mathbf{A}$ from (c) and $\mathbf{b} = [\,5, 0\,]^{\mathsf{T}}$, show that the normal equations have infinitely many solutions, and find them all. Show that $\mathbf{A}^{+}\mathbf{b}$ is one of them, and that it is shorter than the least-squares solution $[\,2, 0\,]^{\mathsf{T}}$.

Part G. Gram–Schmidt and QR

Matrix Unit, §55–§58. Gram–Schmidt turns independent columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ of $\mathbf{A}$ into orthonormal vectors by

$$\mathbf{w}_k = \mathbf{a}_k - \sum_{j=1}^{k-1} (\mathbf{q}_j^{\mathsf{T}} \mathbf{a}_k)\, \mathbf{q}_j, \qquad \mathbf{q}_k = \frac{\mathbf{w}_k}{\|\mathbf{w}_k\|} \qquad (\S55.2).$$

Keeping the numbers $r_{jk} = \mathbf{q}_j^{\mathsf{T}} \mathbf{a}_k$ for $j < k$ and $r_{kk} = \|\mathbf{w}_k\|$ gives the QR **decomposition** $\mathbf{A} = \mathbf{QR}$: $\mathbf{Q} = [\,\mathbf{q}_1 \ \cdots \ \mathbf{q}_n\,]$ has orthonormal columns, so $\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}_n$, and $\mathbf{R}$ is upper triangular with entries r_{jk} and positive diagonal (§56). Then $\mathrm{col}[\mathbf{Q}] = \mathrm{col}[\mathbf{A}]$ (§57), and the least-squares solution of $\mathbf{Ax} = \mathbf{b}$ solves the triangular system $\mathbf{R}\hat{\mathbf{x}} = \mathbf{Q}^{\mathsf{T}}\mathbf{b}$ (§58.1).

Problem G.1 (A QR decomposition, and least squares with it)**.**
Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 0 \\ 2 & 4 \end{bmatrix}.$$

(a) Run Gram–Schmidt on the columns of $\mathbf{A}$: find r_{11}, $\mathbf{q}_1$, r_{12}, $\mathbf{w}_2$, r_{22} and $\mathbf{q}_2$.

(b) Write down $\mathbf{Q}$ and $\mathbf{R}$, and check that $\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}_2$ and $\mathbf{QR} = \mathbf{A}$.

(c) Let $\mathbf{b} = [\,3, 3, 0\,]^{\mathsf{T}}$. Find the least-squares solution of $\mathbf{Ax} = \mathbf{b}$ by solving $\mathbf{R}\hat{\mathbf{x}} = \mathbf{Q}^{\mathsf{T}}\mathbf{b}$ by back substitution. Check your answer against the normal equations.

(d) Find the QR decomposition of the square matrix $\mathbf{C} = \left[\begin{smallmatrix} 3 & -1 \\ 4 & 7 \end{smallmatrix}\right]$. Its $\mathbf{Q}$ is a rotation $\mathbf{R}_\theta$: give $\cos\theta$ and $\sin\theta$. Check that $\det \mathbf{C} = \det \mathbf{Q} \det \mathbf{R}$.

Problem G.2 (What $\mathbf{Q}$ does)**.**
In (a) and (b), $\mathbf{Q} \in \mathbb{R}^{m \times n}$ is any matrix with $\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}_n$.

(a) Prove that $(\mathbf{Qx})^{\mathsf{T}}(\mathbf{Qy}) = \mathbf{x}^{\mathsf{T}}\mathbf{y}$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, and deduce that $\|\mathbf{Qx}\| = \|\mathbf{x}\|$.

(b) Prove that $\mathbf{QQ}^{\mathsf{T}}$ is an orthogonal projection matrix. Then prove that if $m > n$, then $\mathbf{QQ}^{\mathsf{T}} \neq \mathbf{I}_m$.

Hint. D.2(b) bounds the rank of a product.

(c) Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have independent columns, with QR decomposition $\mathbf{A} = \mathbf{QR}$. Explain why $\mathbf{R}$ is invertible, and prove that $\mathbf{A}(\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathsf{T}} = \mathbf{QQ}^{\mathsf{T}}$, naming the laws of §53.1 you use.

(d) For the $\mathbf{Q}$ of Problem G.1, compute $\mathbf{QQ}^{\mathsf{T}}$, and check that $\mathbf{QQ}^{\mathsf{T}}\mathbf{b} = \mathbf{A}\hat{\mathbf{x}}$ for the $\mathbf{b}$ and $\hat{\mathbf{x}}$ of G.1(c).

Questions about any problem here belong in the class Discord, where everyone can see the answer. Questions that are specific to your own work — "is my proof of D.2(b) valid?" — are better brought to office hours, Thursdays 3:00–4:00 PM in Kiely Hall 508. Please email ahead.